

The house or the host?

Inferring superhost status in Melbourne's
short-term rental market, once Airbnb's own
criteria are taken away

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Each dot $\approx$ 66 listings · orange = superhost listing (30.65%)

RESEARCH QUESTION

Can superhost status be inferred from listing attributes and host operating behaviour once Airbnb's own criteria are excluded, and is it more closely associated with listing fundamentals or operational strategy?

THE HOUSE · 3 FEATURES

Listing fundamentals

What the listing is

room_type

dist_cbd

accommodates

THE HOST · 8 FEATURES

Operational strategy

How the listing is run — choices the host makes

days_since_last_review

has_review

availability_365

amenity_count

price

price_missing

minimum_nights

host_listings_count

REMOVING THE SCORECARD

Airbnb's four criteria, and what we did

Superhost criterion	Where it lives in the data	Treatment
≥ 10 completed stays (or 3 totalling 100 nights)	number_of_reviews, ..._ltm, reviews_per_month	Excluded
Overall rating ≥ 4.8	review_scores_rating and sub-scores	Excluded
≥ 90% response rate in 24 h	host_response_rate — 100% empty	Not observable
Cancellation rate < 1%	no column	Not in data

What remains: **11 features** about the listing and the way it is run — none of them is a criterion.

One label, many listings

25,728

listings · 90 columns

14,113

hosts

30.65%

of listings are superhost listings

26.71%

of hosts (3,770) are superhosts

WHY IT MATTERS

The label belongs to the **host**, so all of a host's listings share it.

55.8% of listings share a host with another listing — a random split would let the model recognise hosts.

So every split is grouped by host.

Three steps, each with a measurable effect

01

Review recency

4,477 listings (17.4%) never reviewed: only 6.59% are superhost listings, vs 35.72%.

Kept: recency = 4,594 plus a has_review flag.

REMOVE IT

Tree CV F1 0.575 → **0.526**

02

Amenity count

Comma-splitting the amenity list miscounts 3,096 rows (12.03%), by up to 21 items.

Parsed as JSON instead: one exact count per listing.

REMOVE IT

k-NN CV F1 0.537 → **0.475**

03

Missing price

25.47% of prices missing; 99.85% of unavailable listings have none — MAR, not MCAR.

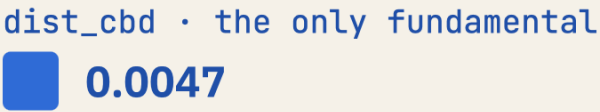
Training-set median (\$245) plus a price_missing flag.

DELETE ROWS INSTEAD

Superhost share 30.65%
→ **38.06%**

Strong on recency, weak on distance

NMI WITH SUPERHOST STATUS

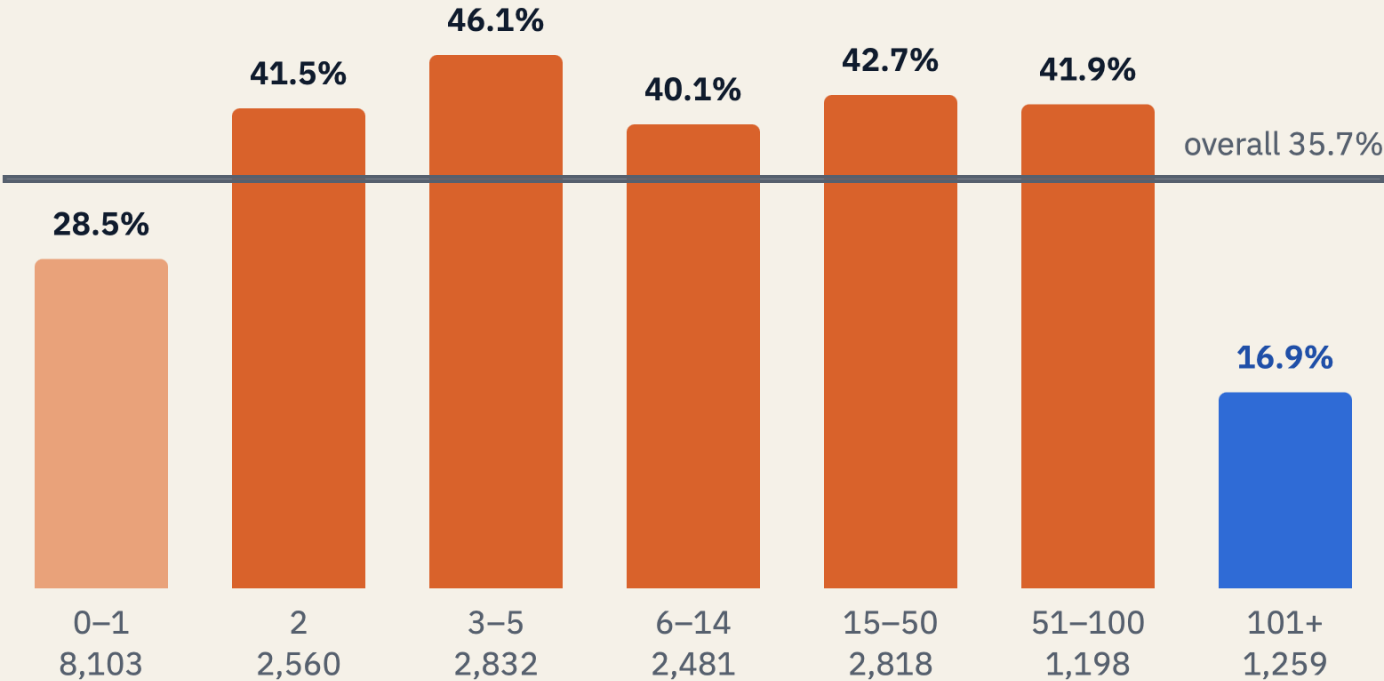


With the target	Pearson	Spearman
recency	-0.337	-0.443
availability	+0.146	+0.155
host listings	-0.069	+0.070
distance	+0.076	+0.009

Pearson on a 0/1 target is point-biserial: it only compares group means. MI and NMI use 10 equal-frequency bins (≈212 listings per cell).

PEARSON -0.069 VS SPEARMAN +0.070

Opposite signs, one curve



SPEARMAN +0.070 · RANKS

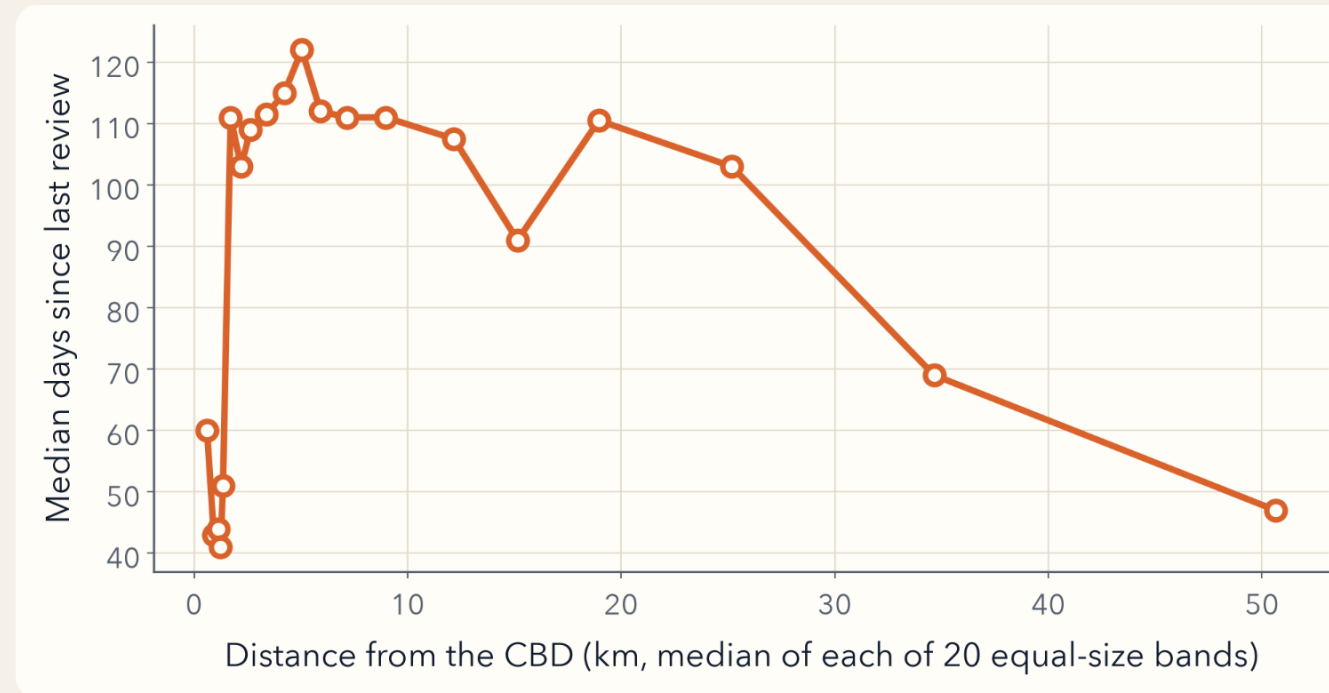
13,495 listings sit in the rising 0–5 range, where the rate climbs 28.5% → 46.1%.

PEARSON -0.069 · RAW VALUES

1,259 listings of hosts with 101+ listings (up to 667) drop to 16.9% and pull the line down.

PEARSON -0.081 VS SPEARMAN $+0.061$

The same pattern, now with distance



Most listings lie on the rising part → **Spearman $+0.061$** ; the few distant, recently reviewed ones pull **Pearson to -0.081** .

INNER CITY · 0.6–1.4 KM

41–60 days

the most recently reviewed listings

MIDDLE SUBURBS · 1.7–5 KM

111 → 122

days: most listings sit on this rise

FAR OUT · 34.6–50.6 KM

69 → 47

days: few listings, recently reviewed

No host is ever seen twice

25,728 listings

grouped by host · stratified on the label

TRAIN 80%

20,582

11,292 hosts · 5-
fold grouped CV

TEST 20%

5,146

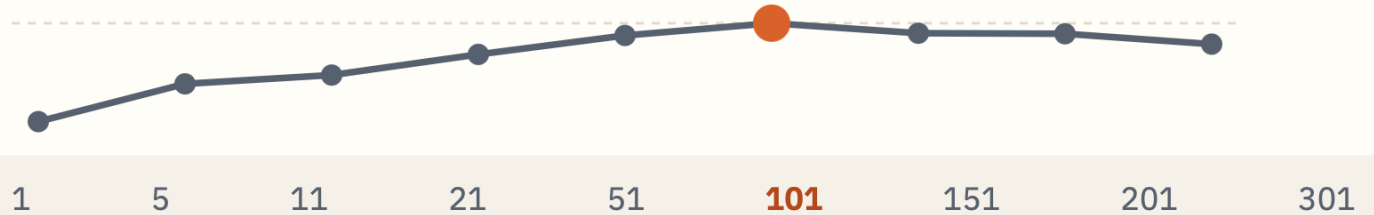
2,821 hosts
· used once

METRIC: SUPERHOST-CLASS F1

Always guessing “not a superhost” scores
0.694 accuracy but F1 = 0.

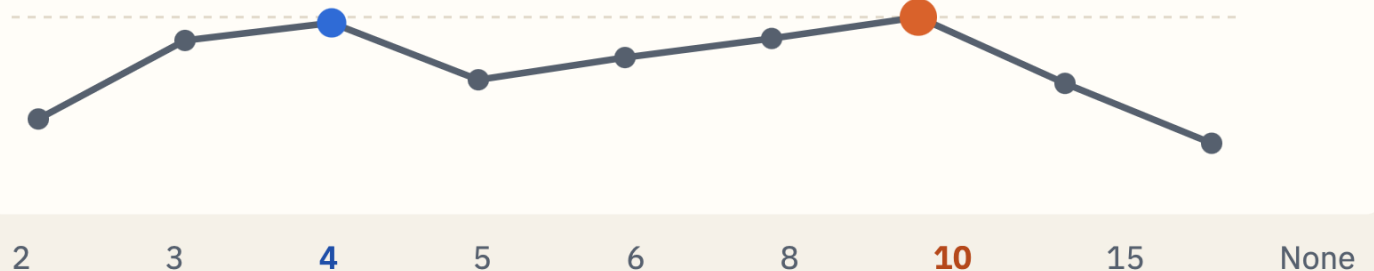
k-NN · CV F1 by k

k = 101 · 0.537



Decision tree · CV F1 by max_depth

depth 10 · 0.5745



Both curves share one y-scale (0.47–0.59) · depth 4 (blue) is within one SD of depth 10: 0.5712 vs 0.5745

TEST SET · 5,146 LISTINGS · 1,577 SUPERHOST LISTINGS

Two in three superhost listings found

DECISION TREE · DEPTH 10

0.632

F1 · 95% CI [0.594, 0.676]

recall 0.664 · 1,047 of 1,577

K-NN · K = 101

0.577

F1 · CI [0.542, 0.616]

recall 0.560 · 883 found

MAJORITY BASELINE

0.000

F1 · accuracy 0.694

finds none of them

TREE – K-NN

+0.055 F1, CI [0.013, 0.099] — excludes zero

GAIN OVER BASELINE

F1 +0.632 · macro-F1 +0.319 · accuracy +0.069

Where each model goes wrong

		k-NN · k = 101		Decision tree · depth 10	
Actual	Not superhost	2,970	599	2,879	690
	Superhost	694	883	530	1,047
		Not superhost	Superhost	Not superhost	Superhost
		Predicted		Predicted	

SUPERHOSTS FOUND

1,047 vs 883

true positives · tree vs k-NN

THE PRICE: FALSE ALARMS

690 vs 599

non-superhosts flagged · tree vs k-NN

STILL MISSED

530 vs 694

superhost listings not found

The tree labels **1,737** listings as superhosts against **1,482** for k-NN: it finds more, and pays with more false alarms.

THE DECISIVE EXPERIMENT · TRAIN ON ONE FAMILY ONLY

The signal lives in how the listing is run

Listing fundamentals only · 3 features

 **0.196**

Operational strategy only · 8 features

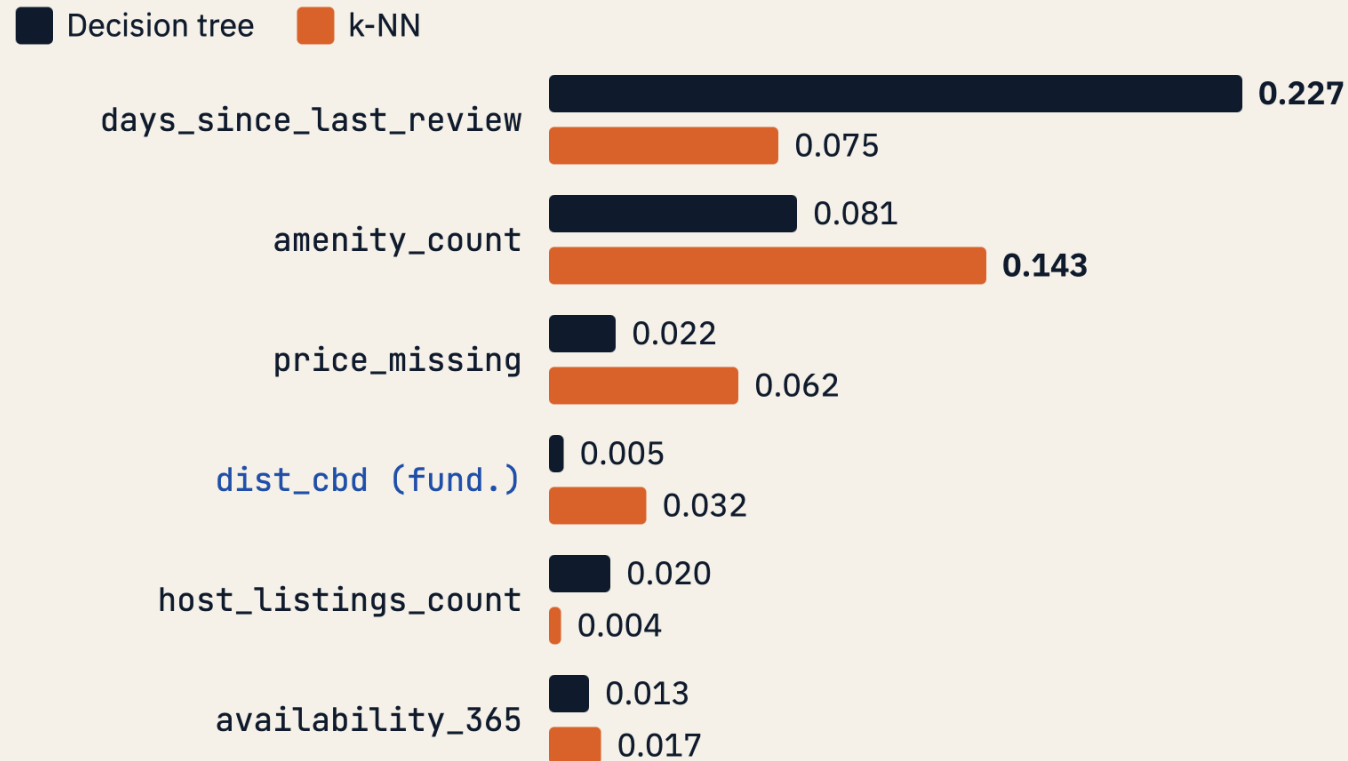
 **0.639**

GAP
+0.443 · CI [0.381, 0.505]

NO REVIEW FEATURES
still 0.566 · +0.371
over fundamentals

PRICE MOVED TO FUNDAMENTALS
CV 0.184 vs 0.558 · same verdict

Same family, different favourites



TOP 3 · FILTER (MI)

recency · host listings · **availability**

TOP 3 · EMBEDDED (TREE)

recency · host listings · **amenities**

FILTER HIGH, TREE ZERO

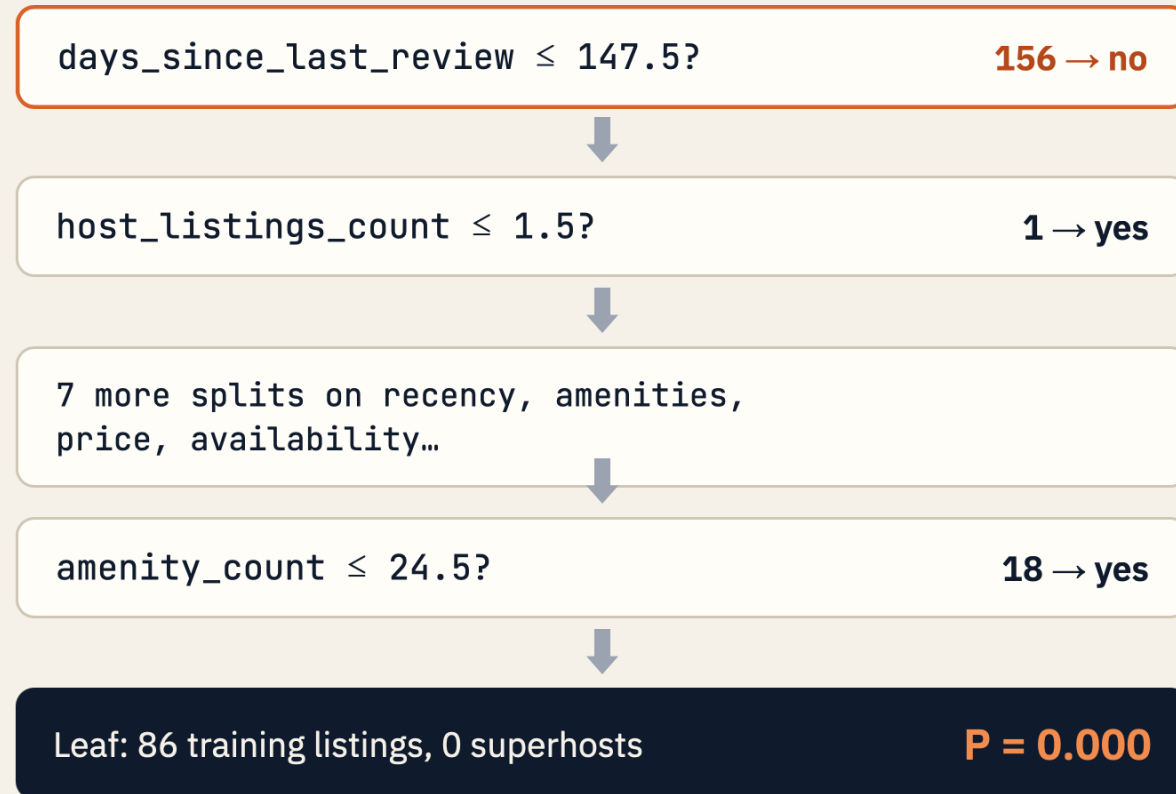
has_review: MI rank 7, impurity 0 — it is simply “recency = 4,594”.

DROP AVAILABILITY_365 (EXP. E)

CV F1 0.5846 vs 0.5745 · test 0.630 vs 0.632

A HARD CASE · THE TREE'S MOST CONFIDENT MISS

Listing 6812677: certain, and wrong



WHAT THE MODEL COULD NOT SEE

24 reviews in the last 12 months · needs 10
rating 4.8 · needs 4.8

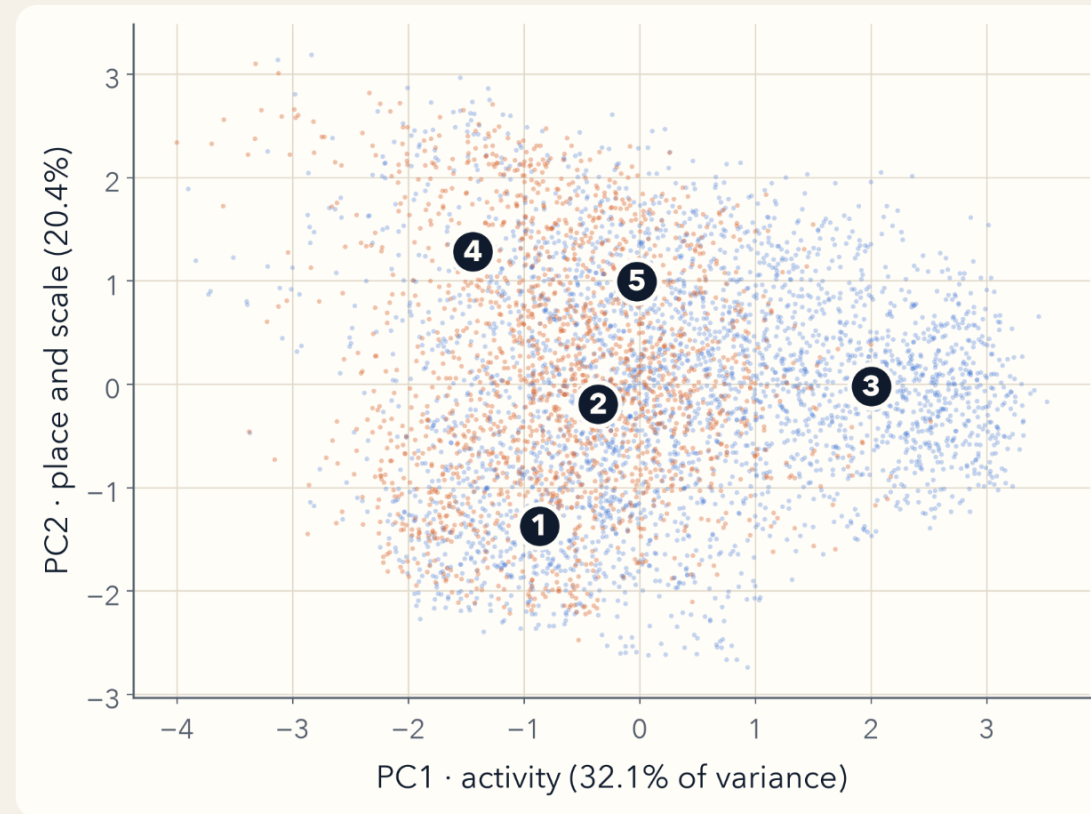
Meets both criteria visible in the data — both excluded by design.

AN OUTLIER WITHIN ITS CLASS

Recency of 156 days: 54.6th percentile of all listings,
90.8th among reviewed superhosts.

k-NN misses it too: P = 0.267.

Variance is not relevance



● Not superhost ● Superhost ● K-Means centroid

PC1 · activity axis

32.1%

recency +0.563 · amenities -0.504 ·
availability -0.447

with superhost status: $r = -0.354$

PC2 · place and scale

20.45%

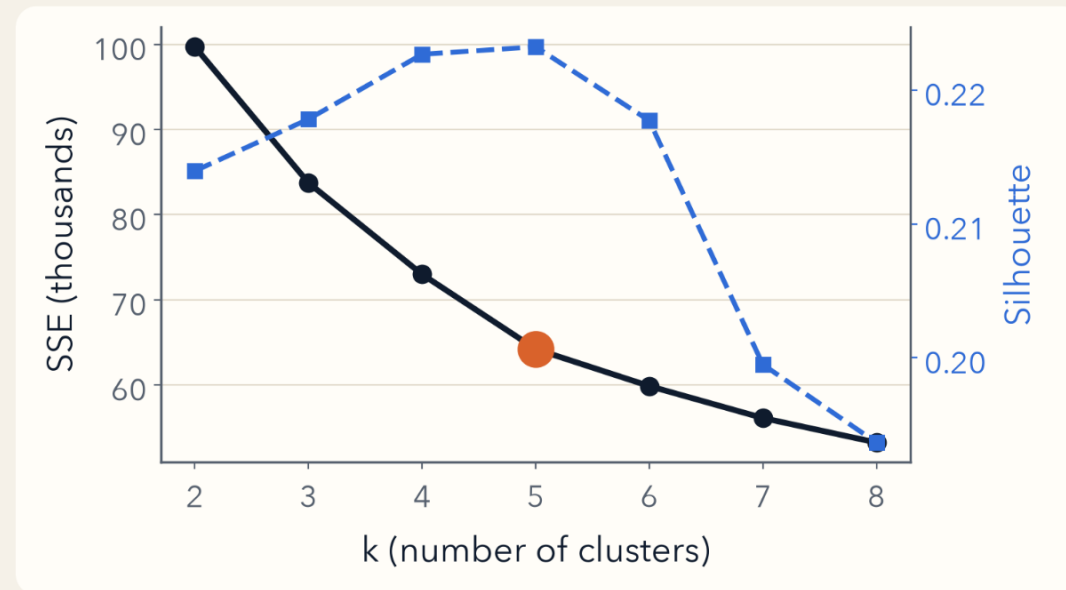
distance +0.786 · host listings -0.556 ·
capacity +0.208

with superhost status: $r = +0.050$

One fifth of the variance, almost none of the signal:
the label sits on the operational axis.

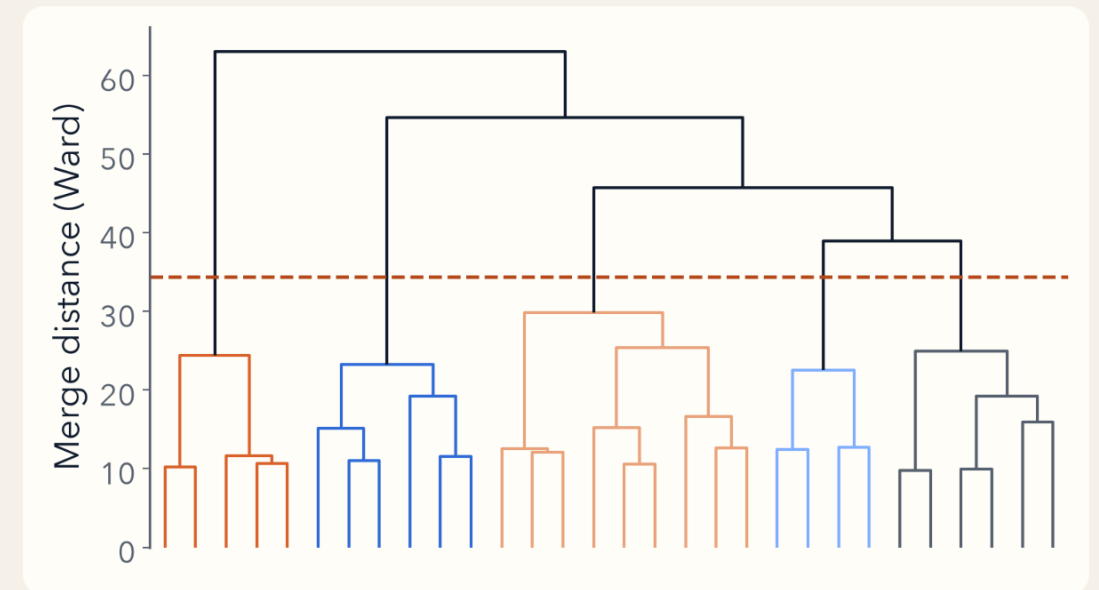
Choosing $k = 5$ and Ward linkage

ELBOW AND SILHOUETTE · $K = 2$ TO 8



SSE falls 12.0% from $k = 4$ to 5 , then only 6.8%; silhouette peaks at $k = 5$ (0.2232 vs 0.2227 at $k = 4$).

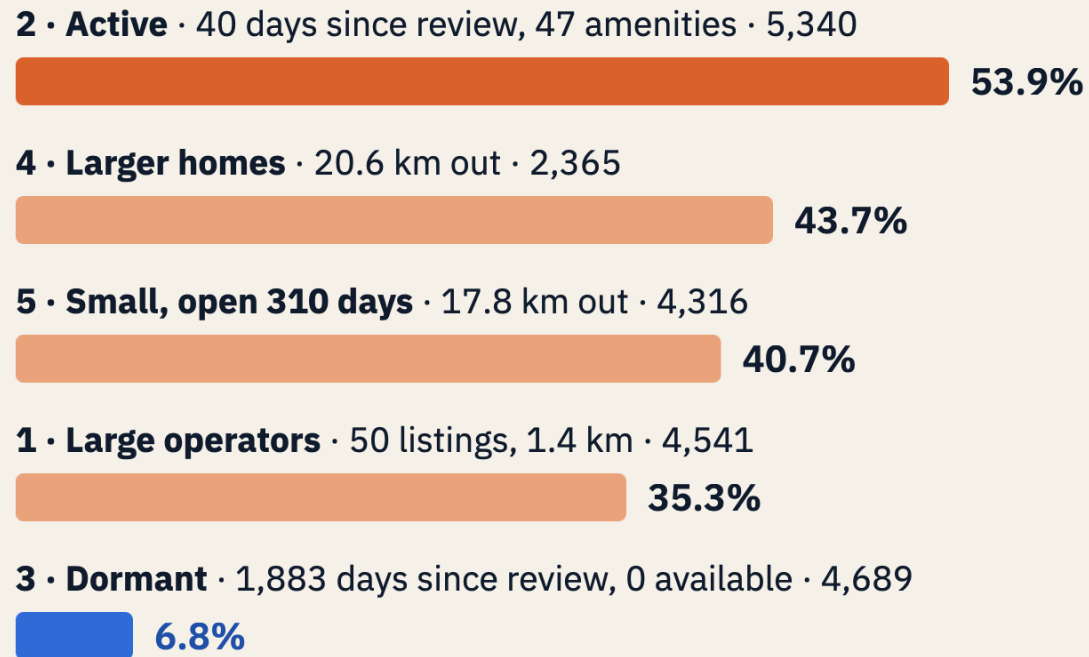
WARD DENDROGRAM · 2,000-LISTING SAMPLE



Single linkage leaves 21,247 of 21,251 listings in one cluster; average and complete also collapse. Ward minimises SSE, like K-Means.

Activity splits the market, not distance

K-MEANS CLUSTERS · SUPERHOST RATE



K-MEANS VS WARD

ARI 0.377

Same structure, different boundaries.

Dormant group: 99.2% of Ward's dormant cluster is K-Means cluster 3.

Active group: 82.6% of K-Means cluster 2 lands in Ward W1.

Fundamentals only: superhost-rate range across clusters just 0.075.

LIMITATIONS

What we would test next

Host label, listing rows

Rows are not independent. Next: one row per host, 14,113 rows.

Recency as a proxy

$\rho = -0.82$ with reviews in the last 12 months. Without it, operations still lead by 0.371.

Depth within noise

Depth 10 vs 4: 0.0033 apart, under one SD. Train F1 0.719 vs test 0.632.

Unequal families

8 operational vs 3 fundamental features; part of the gap may be set size.

A narrow k

Silhouette 0.2232 at $k = 5$ vs 0.2227 at $k = 4$. A gap statistic would settle it.

Imputed prices

93.35% of missing prices belong to unavailable listings; the median hides them.

THE ANSWER

Partly — and it is the host, not the house.

0.639

test F1 from operational features alone,
against 0.196 for fundamentals

0.632

best F1 with the criteria removed; 1,047
of 1,577 found

118 / 124

superhost listings inactive for over a
year that the tree missed

Thank you — questions?